Supply-Aware Sequential Matching: Round 2 Technical Report

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Challenge: Vouchsafe × IITM, Sequential Matching Problem (kit 1.0.0). Synthetic simulator only; nothing here is evidence about real people. Primary outcome: msmi per 100 arrived members, reported with calibration and who is left waiting.

1 Summary

Round 2 asked three questions. (1) The stdlib build reproduces the Round 1 prototype's decisions and, within noise, its results. (2) Shrunk member propensities, with priors recalibrated on selection seeds, do not show a reliable gain over a constant prior on fresh seeds; their ranking signal is weak (AUC 0.55, Brier equal to the base rate). (3) Under changed missingness and response behaviour, targeted constraint asks remain essential, soft-field asks never pay, and learning is at most weakly useful; the stress runs have only 4 episodes per condition, so this part is suggestive. The one change that held on both seed sets is a *waiting priority*: it reduces members left waiting and waiting time without a detectable change in msmi, mutual acceptances or dates against the supplied greedy baseline. It does trade away the mutual-acceptance gain that the Round 1 build showed pooled.

2 What changed

Round 2 default (`r2`). Every member has the same prior (no learned propensities); the daily batch is an exact maximum-weight matching (components ≤ 18 members, greedy beyond); an edge weight is multiplied by $1+\lambda(s_a+s_b)$ with $\lambda = 2$, where $s_i = \min(\text{days since arrival}, 30)/30$ for members never yet introduced and 0 otherwise; matching relationship_goal (when both are observed) multiplies by 1.5; hard-constraint asks are targeted; no soft asks. **Round 1 build (`r1`):** learned Beta-shrunk propensities, no waiting term; kept as a profile. Hard eligibility always comes from the supplied `kit.eligibility`; outputs are re-validated and fall back to the supplied baseline on any error. The waiting term is my own definition; λ was not tuned beyond a sweep.

3 Protocol

All comparisons are paired on identical worlds, in-process, with the unmodified simulator (an exact memo of `eligibility` only speeds it up). *Stage A* (selection): seeds 3001–3009 × 6 scenarios = 54 episodes. *Stage C* (fresh, untouched during selection): seeds 6001–6006 × 6 = 36. *Stress*: seeds 7001–7004 in five modified worlds (70% of answers removed and 15% declined; response rates ×0.6; response rate halved at day 30; bimodal response rate; strong member acceptance bias), 20 episodes. Intervals are 95% percentile bootstraps over paired episode differences (2000 resamples). "Left waiting" counts members with a truly feasible partner who never received an introduction; it uses simulator truth after the fact and is a diagnostic only. Calibrated priors were fitted by beta-binomial maximum likelihood on stage-A feedback only: responds $(0.78, 12)$, Yes $(0.45, 8)$, second-meeting Yes $(0.66, 50$, capped) versus hand-set $(0.8, 4), (0.4, 4), (0.53, 3)$.

4 Results

Q1: does the build reproduce the prototype? On fresh seeds the prototype (networkx blossom, soft asks) versus the stdlib Round 1 build differs by -0.28 msmi $[-0.69, +0.14]$ and $+0.22$ mutual acceptances $[-0.44, +0.97]$, within noise; it does spend $+53.7$ ask units and gets $+0.89$ dates $[+0.17, +1.67]$ (one of eight outcomes, not corrected for multiplicity). At the decision level, on 337 days from 9 fresh episodes the exact stdlib matching has the same total weight as networkx on 333 days (98.8%) and the identical pair set on 285

Policy (vs greedy)	Left waiting	Mean wait (d)	msmi/episode	Mutual acc.	Dates
Round 1 build	-0.30 $[-0.60, 0.00]$	-0.80 $[-0.99, -0.60]$	$+0.11$ $[-0.16, +0.38]$	$+0.66$ $[+0.02, +1.31]$	$+0.61$ $[-0.10, +1.40]$
Constant prior, no waiting	-0.23 $[-0.58, +0.12]$	-0.79 $[-1.00, -0.59]$	$+0.11$ $[-0.12, +0.34]$	0.00 $[-0.74, +0.76]$	-0.08 $[-0.81, +0.77]$
Round 2 (waiting)	-1.24 $[-1.52, -0.96]$	-1.00 $[-1.18, -0.81]$	$+0.08$ $[-0.19, +0.37]$	-0.07 $[-0.77, +0.63]$	-0.06 $[-0.83, +0.76]$
Waiting + learned	-1.27 $[-1.54, -0.99]$	-0.95 $[-1.13, -0.77]$	$+0.21$ $[-0.06, +0.47]$	$+0.09$ $[-0.53, +0.71]$	$+0.22$ $[-0.51, +0.96]$
Waiting, greedy alloc.	-1.14 $[-1.42, -0.88]$	-0.96 $[-1.14, -0.79]$	-0.07 $[-0.27, +0.13]$	-0.09 $[-0.60, +0.44]$	$+0.31$ $[-0.27, +0.87]$
Waiting, baseline asks	-1.00 $[-1.27, -0.73]$	-0.23 $[-0.36, -0.10]$	$+0.14$ $[-0.03, +0.32]$	$+0.01$ $[-0.59, +0.60]$	$+0.08$ $[-0.50, +0.66]$
Waiting + soft asks	-1.20 $[-1.48, -0.92]$	-1.02 $[-1.21, -0.84]$	$+0.01$ $[-0.24, +0.30]$	-0.32 $[-1.04, +0.42]$	-0.29 $[-1.06, +0.53]$

Table 1: Pooled stages A+C (90 paired episodes), difference from the supplied greedy baseline. Ask cost: -3 units for targeted asks, 0 for baseline asks, $+51.5$ for soft asks. Greedy's mean is about 0.7 msmi per episode.

(84.6%; ties). The largest value gap is 5.8%, and exactly four days had a component above the 18-member exact limit, as many as the unequal days.

Q2: do calibrated propensities help? Hand-set predictions of mutual acceptance average 0.10 against 0.137 observed; fitted priors give 0.118. On fresh seeds the Brier score is 0.119 (hand-set) and 0.118 (fitted), identical to predicting the base rate (0.118); AUC is 0.554 and 0.549 (0.52 on stage A). Fitted-prior versus constant-prior outcomes on fresh seeds: msmi $+0.08$ $[-0.17, +0.31]$, mutual acceptances $+0.56$ $[-0.14, +1.28]$. Hand-set learned versus constant prior, pooled: msmi $+0.13$ $[+0.02, +0.26]$, mutual acceptances $+0.16$ $[-0.28, +0.59]$, but fresh seeds alone gave $+0.17$ $[-0.06, +0.42]$ msmi and the best-looking variant was chosen among many, so I do not call it confirmed. An oracle using the simulator's true acceptance probabilities gets $+1.03$ mutual acceptances $[+0.25, +1.81]$ over greedy on fresh seeds but no gain on waiting, so ordering signal exists but is small relative to what 1–3 observations per member can learn.

Q3: changing conditions. With only 4 episodes per condition, intervals are wide. Pooled over the 20 stress episodes the waiting rule still lowers members left waiting (-0.30 $[-0.55, -0.05]$ vs greedy), and targeted constraint asks remain essential: removing all asks adds $+37.9$ $[+30.7, +44.4]$ left-waiting members and loses 5.0 mutual acceptances versus greedy. Soft asks cost about $+45$ units for $+0.05$ msmi $[0.00, +0.15]$ over the default and never helped. Learning on top of the default is $+0.05$ msmi $[-0.10, +0.20]$ overall and $+0.25$ $[0.00, +0.75]$ in each of the two conditions where members really differ (bimodal response, strong acceptance bias), the only place a gain is plausible; when response behaviour drops mid-episode it is -0.25 $[-0.75, 0.00]$. Baseline asks give $+0.65$ mutual acceptances versus the default for $+2.6$ units, again within noise. The 70%-missing world is degenerate (greedy coverage 0.04).

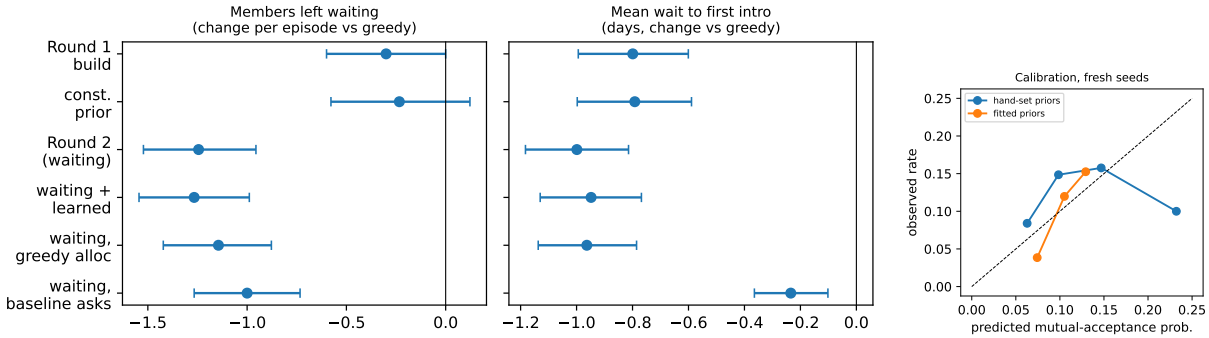


Figure 1: Left: paired differences from greedy (pooled A+C, 95% intervals). Right: reliability of the mutual-acceptance prediction on fresh seeds.

5 What the comparisons taught

(i) The waiting priority does the work: exact matching adds nothing over greedy allocation with the same priority on who is left waiting (-1.24 vs -1.14), so a simpler allocator would do; I keep exact matching because it costs 0.13 s per call. (ii) Against the Round 1 build, the waiting rule is a trade: pooled, Round 1 has $+0.72$ $[+0.23, +1.20]$ mutual acceptances and $+0.67$ $[+0.20, +1.16]$ dates over Round 2 but leaves 0.94 $[0.67, 1.23]$ more members waiting; Round 2 is only "free" relative to greedy. (iii) The waiting+learned variant recovers most of both but did not beat the default on fresh seeds alone; it is the candidate for a properly powered Round 3. (iv) Targeted asks are mainly a waiting-time tool (mean wait -1.00 vs -0.23 with baseline

asks), not a cost saver (−3 units). (v) msmi stays too rare (under one event per episode) to separate any of these; mutual acceptance, dates and waiting are the usable outcomes.

6 Limitations

Stage-A selection used about eleven policies, so its single positive msmi results are subject to selection; stage C and the pooled intervals are uncorrected for multiplicity. Stress runs have 4 episodes per condition. $\lambda = 2$ is an untuned choice (1, 2, 4 gave the same left-waiting result on stage A). The official evaluator was run for seed 101 only (all six scenarios valid; primary score 0.50, coverage 0.4175, mutual acceptances 8.5 per 100, ask cost 199, 8.3 s inference in total); the Docker container build was not run in my environment.

7 Reproducibility and provenance

Python 3.12, standard library only for the policy; analysis used numpy, scipy, matplotlib and networkx (prototype only). No training; priors fixed constants or fitted on seeds 3001–3009; deterministic. See `README_SUBMISSION.md` and `run_round2.sh`. The simulator source was read to design experiments and the oracle uses simulator truth only inside the evaluation harness. Coding and analysis assistance: Claude (Anthropic); every reported number comes from the included result files.